

Experiment 38

Aim

To detect human faces in an image using OpenCV and mark the detected faces with rectangular bounding boxes.

Procedure

1. Import the OpenCV library.
2. Load the input image.
3. Load the pre-trained Haar Cascade face detection classifier.
4. Convert the image into grayscale.
5. Detects faces using detectMultiScale().
6. Draw rectangles around the detected faces.
7. Display and save the output image.

Code:

```
import cv2
img = cv2.imread("image.jpg")
if img is None:
    print("Error: Image not found!")
    exit()
face_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades + "haarcascade_frontalface_default.xml"
)
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
faces = face_cascade.detectMultiScale(
    gray,
    scaleFactor=1.1,
    minNeighbors=5,
    minSize=(30, 30)
```

```

)
for (x, y, w, h) in faces:
    cv2.rectangle(
        img,
        (x, y),
        (x + w, y + h),
        (0, 255, 0),
        2
    )
print("Number of faces detected:", len(faces))
cv2.imshow("Face Detection", img)
cv2.imwrite("face_detected.jpg", img)
cv2.waitKey(0)
cv2.destroyAllWindows()

```

Sample Input:



Output:

```

>>> |===== RESTART: C:/Users/navya/Downloads/CV/Exp 38.py =====
      |C:\Users\navya\AppData\Local\Programs\Python\Python312\Lib\site-packages\cv2\__init__.py
      |4.10.0
      |Number of faces detected: 0

```

Result

The faces in the given image were successfully detected using OpenCV. Rectangular bounding boxes were drawn around the detected faces, and the output was saved as face_detected.jpg.